Inequalities — problem 7.70

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A function $d(x, y)$ of two points is a *distance* if $d(x, y) = d(y, x)$, $d(x, y) + d(y, z) \leq d(x, z)$ for all x, y, z, and $d(x, x) = 0$. The second property is called *triangle inequality*. Prove that the following function is a distance :

$$d(x, y) = \frac{|x - y|}{\sqrt{1 + x^2}\sqrt{1 + y^2}}.$$